

Public Web & Vendor Onboarding Specification

apps/web: 3D WebGL Canvas, 10-Step Supplier Qualification Train & Marketing Portal

APPS/WEB

Project: RVCC Enterprise Monorepo	Version: 2.0.0 (Production)	Architecture: Next.js 16 + Edge + Prisma	Status: Verified & Active
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1. Application Overview & Objectives

`apps/web` is the primary public-facing corporate website, marketing portal, and vendor self-service onboarding hub for RVCC. It delivers high-impact visual presentation through 3D architectural rendering while providing high-performance interactive tools for clients, job applicants, and prospective suppliers.

Target Audience & Portals	Core Technology Stack
Corporate Clients: Explore civil engineering, electromechanical, landscaping, and infrastructure project portfolios. Prospective Vendors: Multi-step audited supplier registration train. Job Applicants: Interactive careers listing and resume submission portal. Stakeholders: Interactive PDF flipbook for company profiles and quality policy.	Framework: Next.js 16 (App Router + Turbopack). Rendering Engine: Three.js / WebGL with custom GLSL shaders. Styling & Animations: Tailwind CSS, Lenis Smooth Scroll, Framer Motion. State Management: React Context + LocalStorage Draft Persistence.

2. Route Tree & Page Topology

Route Path	Rendering Mode	Key Components & Functionality
<code>/</code> (Home)	Static Prerender (SSG)	Hero, Skyscraper 3D Canvas, About Overview, Our Works Marquee, Services Carousel, Major Projects showcase, CSR Initiatives.
<code>/about</code>	Static Prerender (SSG)	Company history timeline, Mission/Vision/Values, Executive Leadership cards, ISO Certifications grid, Safety & Sustainability metrics.
<code>/services/[slug]</code>	Incremental Static (ISG)	Dynamic service showcase (Artificial Grass, Architectural Services, Artificial Lakes, etc.) with pre-generated static params.
<code>/projects/[slug]</code>	Dynamic Server / SSR	Interactive project inspection, high-resolution photo galleries, technical specifications, client metadata, and location tags.
<code>/documents</code>	Static + Dynamic	Document showcase with secure PDF Flipbook Reader (<code>FlipbookReader.tsx</code>) and authenticated unlocking gateway.
<code>/careers</code>	Dynamic SSR + SWR	Live job opening board with category filtering, detailed job descriptions, and multi-part resume application upload endpoint.
<code>/enquire/[step]</code>	Dynamic Client Train	10-step wizard for supplier pre-qualification, auto-save state recovery, file attachment processing, and OTP email verification.

3. 3D Architectural WebGL & Interactive Graphics

The web application features high-performance WebGL graphics powered by Three.js:

- `SkyscraperCanvas.tsx`: Real-time 3D rendered skyscraper model with interactive camera orbit controls and dynamic lighting that reacts to user mouse movement.
- `Interactive3DCard.tsx`: CSS 3D perspective transform cards providing tactile depth, specular highlights, and smooth physics-based tilting.
- `LogoMarquee.tsx` & `3d-marquee.tsx`: Infinite hardware-accelerated logo ticker with zero layout shifts and seamless looping.
- `LenisProvider.tsx`: Virtual smooth inertia scrolling delivering a unified luxury aesthetic across desktop and mobile devices.

4. Multi-Step Supplier Onboarding Engine (10-Step Wizard)

The vendor onboarding workflow (`/enquire`) is structured as a fault-tolerant, state-preserving wizard that allows suppliers to complete qualification in stages.

Step #	Step Name	Form Schema & Collected Attributes
1	Company Profile	Legal Trade Name, Commercial Registration (CR) Number, VAT Identification Number, Establishment Date, Website URL.
2	Physical Addresses	Headquarters Location, Branch Offices, City, Postal Code, Building Number, GPS Coordinates.
3	Classifications	Primary & Secondary Business Sectors, General Contracting, Specialized MEP, Landscaping, Material Supply categories.
4	Bank Account	Beneficiary Name, Bank Name, IBAN Number, SWIFT / BIC Code, Account Currency (SAR/USD).
5	Key Contacts	Authorized Executive Representative, Procurement Officer, Finance Contact (Full Name, Designation, Official Email, Phone).
6	Questionnaire	Compliance disclosures, Quality certifications (ISO 9001/14001/45001), Safety Track Record, Annual Turnover.
7	Attachments	CR Certificate PDF, VAT Registration PDF, Chamber of Commerce Certificate, Company Profile deck, Bank Proof letter.
8	Review Summary	Consolidated preview of all 7 sections with inline edit triggers and validation status flags.
9	Email OTP Verification	Cryptographic 6-digit one-time passcode verification to prove ownership of the authorized corporate email domain.
10	Submission & Confirmation	Generates a unique Tracking Reference Number (<code>REG-XXXXXX</code>), triggers backend notification, and displays tracking badge.

State Persistence & Draft Recovery

`EnquireContext.tsx` automatically serializes the form state to encrypted LocalStorage and syncs draft checkpoints to `/api/enquire/draft`. If a supplier accidentally closes their browser, their progress is instantly restored upon re-opening the page.